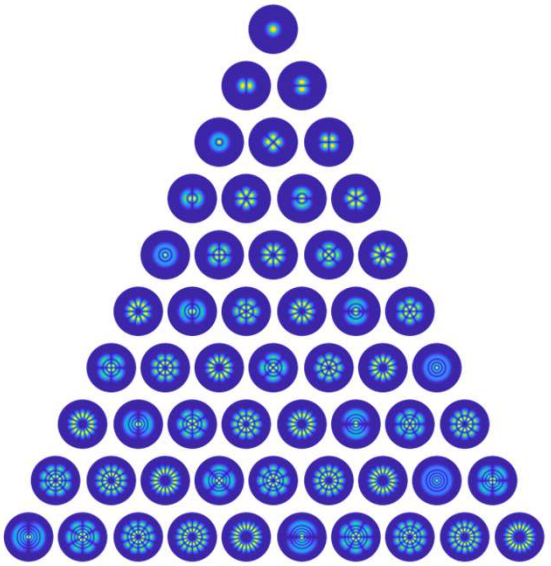
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Advanced optical fiber communications

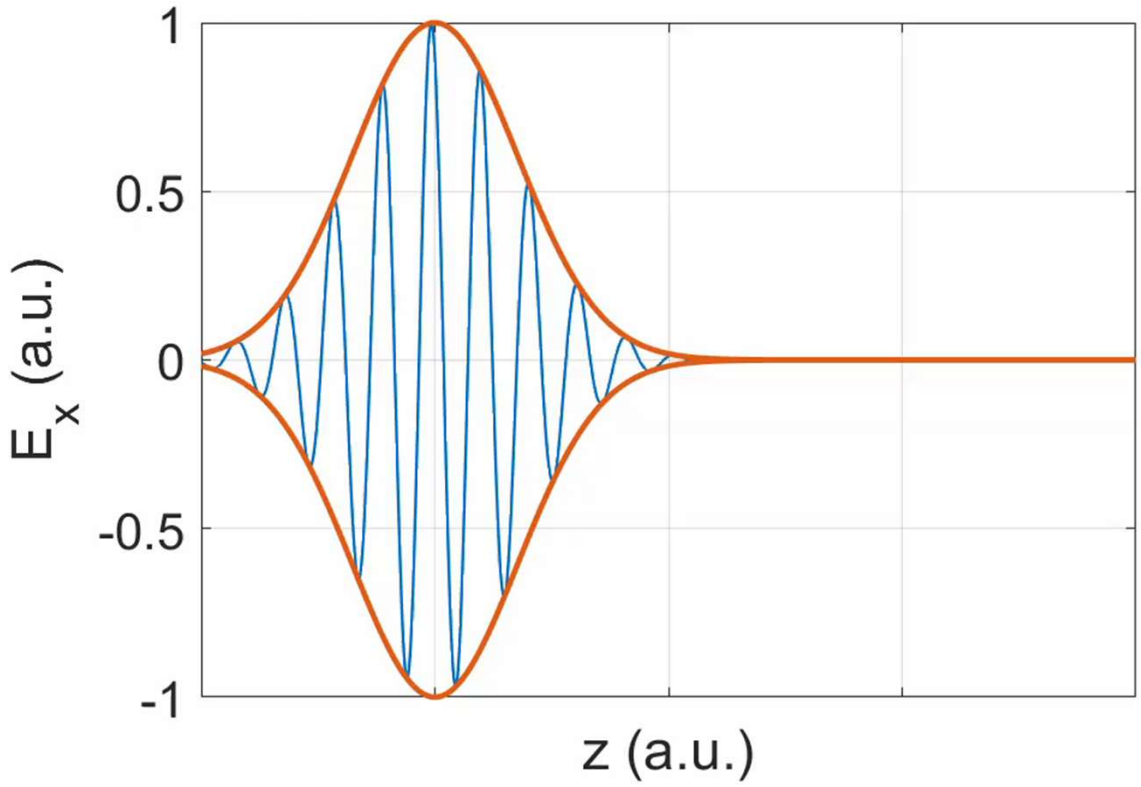
Wintersemester 2025/26

Lecture 3



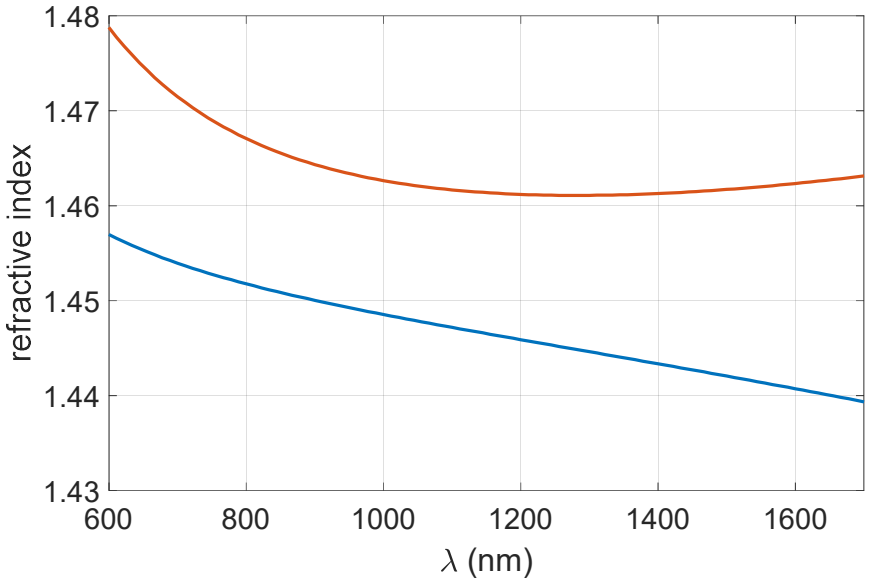
Prof. Dr.-Ing. Georg Rademacher

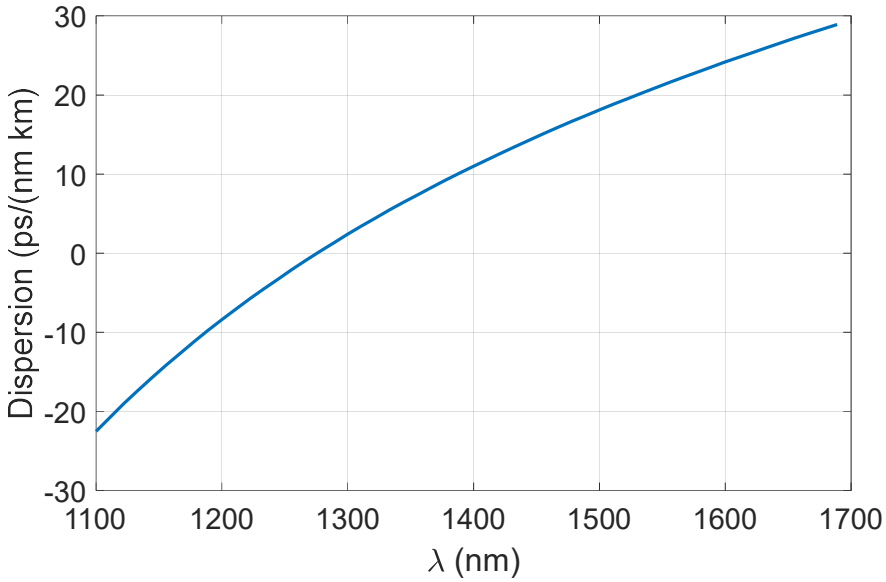
Chromatic Dispersion



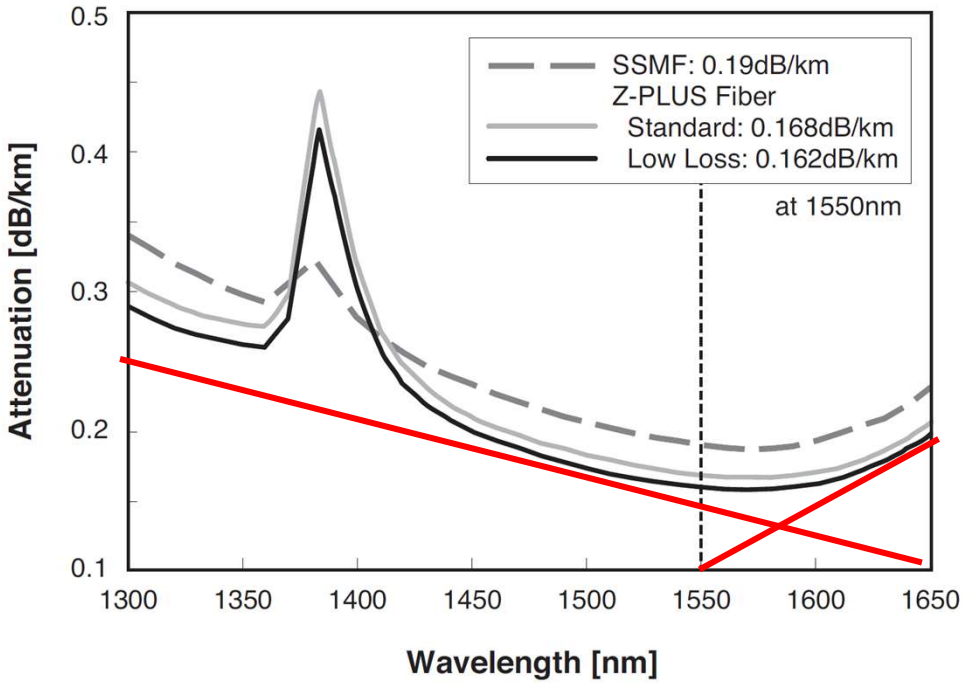
Chromatic Dispersion

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Attenuation



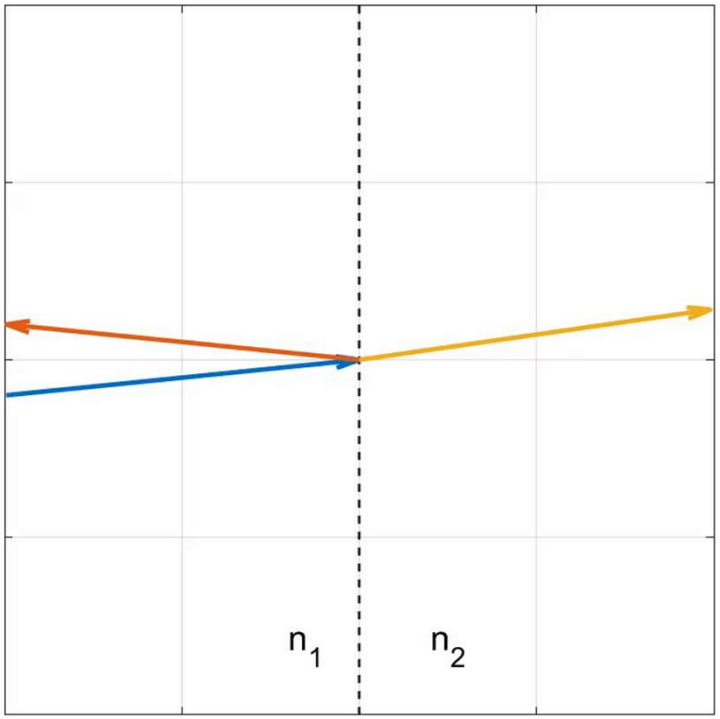
Yamamoto et al, Low-Loss and Low-Nonlinearity Pure-Silica-Core Fiber for Large Capacity Transmission, SEI Technical review, 2013

Plane waves on dielectric interfaces



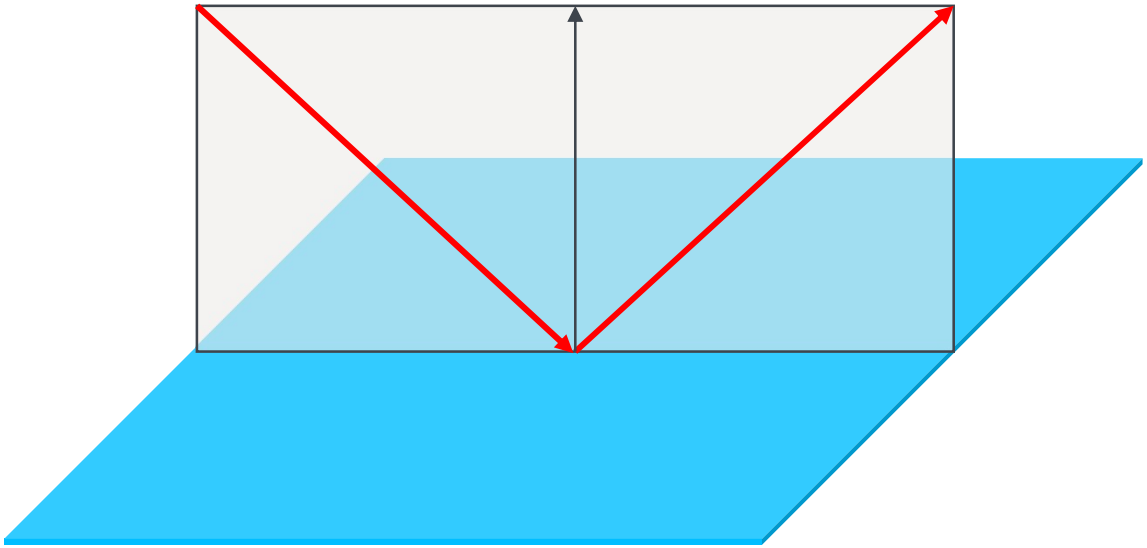
Snell's law



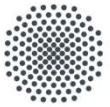


Plane of insidence

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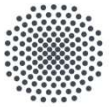


Plane waves on dielectric interfaces



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Plane waves on dielectric interfaces



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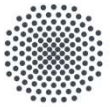
Reflectivity of plane waves on dielectric interfaces



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Plane waves on dielectric interfaces

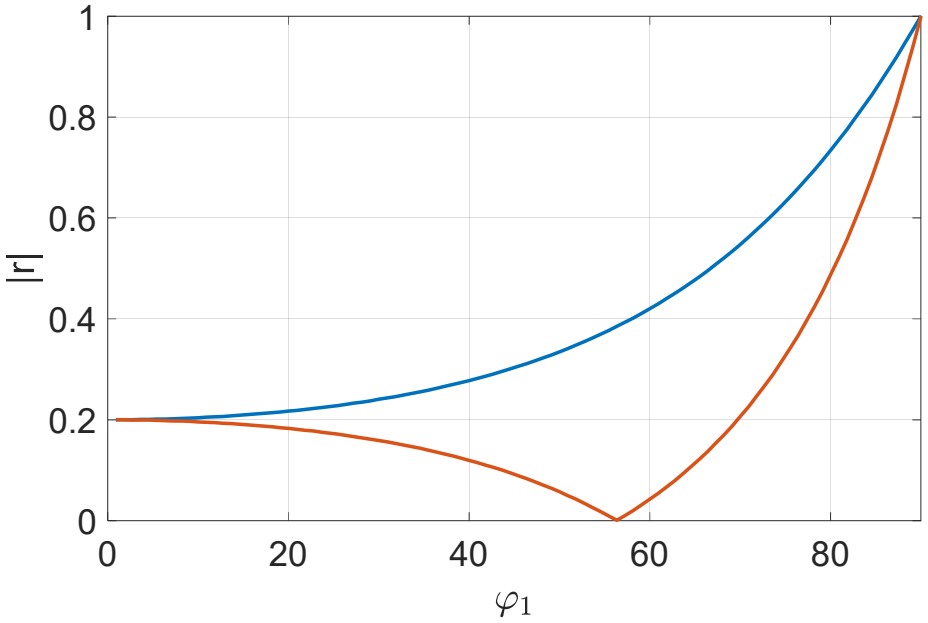
Plane waves on dielectric interfaces

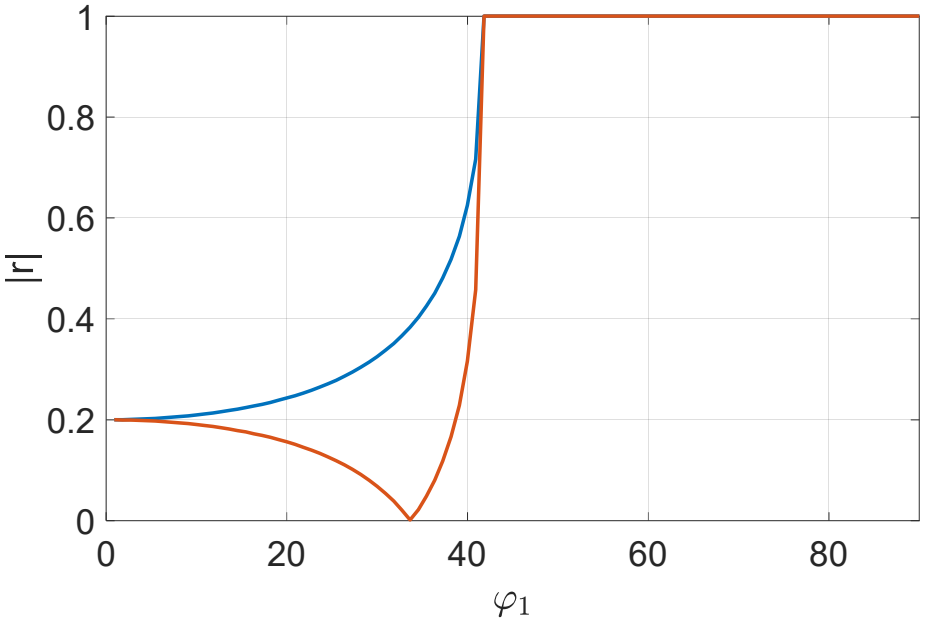


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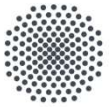
Plane waves on dielectric interfaces

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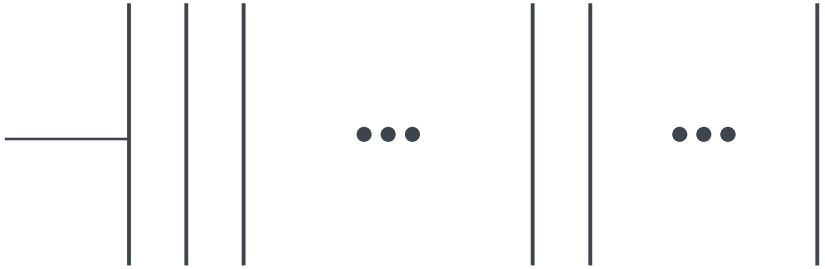
Plane waves on dielectric interfaces



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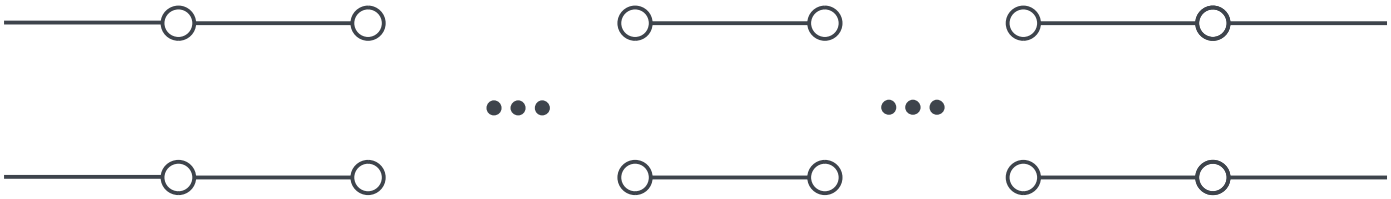
Reflection on multiple layers

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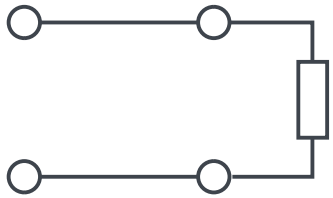


Reflection on multiple layers

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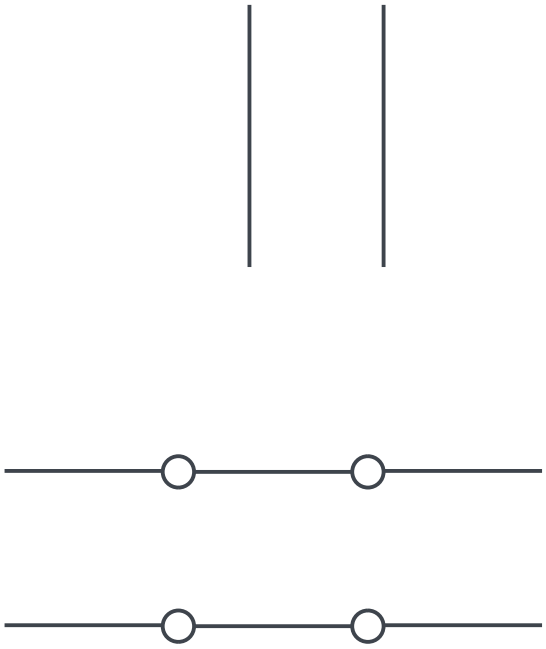


Reflection on multiple layers



Example: $\lambda/4$ layer for anti-reflection coatings

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Example: $\lambda/4$ layer for anti-reflection coatings



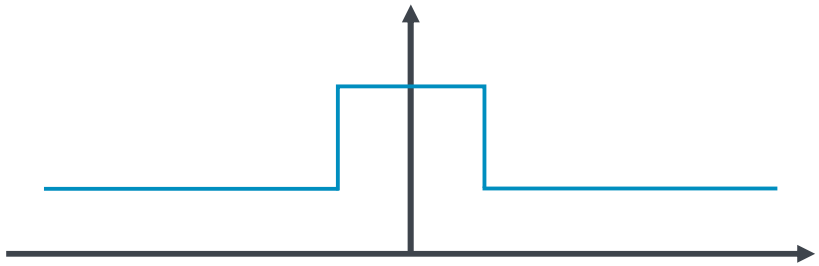
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Step-index fiber

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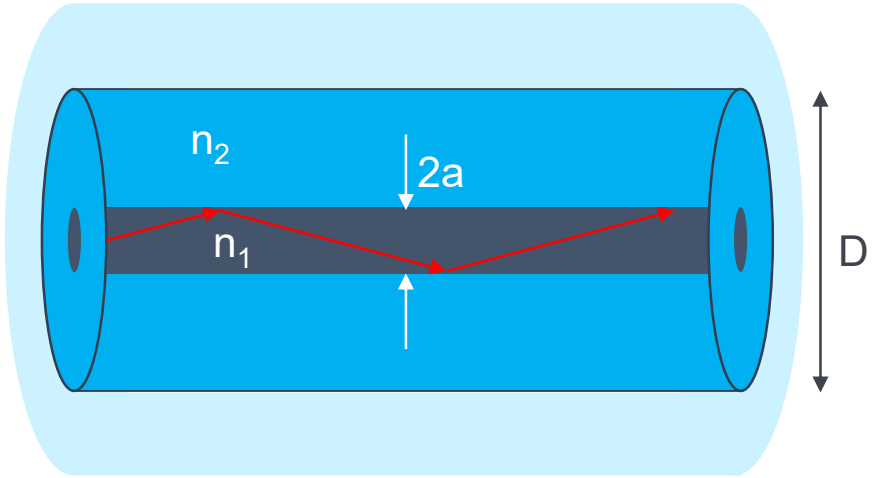


Step-index fiber



Step-index fiber

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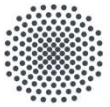
Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

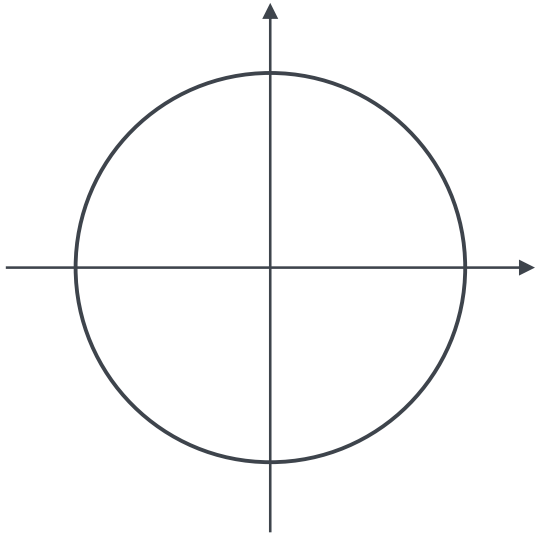
Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber



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Boundary conditions in step-index fiber



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Thank you!



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